

PropertyScope NSW

Release 1 / Technical report

41026 Advanced Software Development — Assessment 2

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Evidence distinguishes real provider answers, deterministic decisions against live services, and isolated tests. Repository links identify the selected report baseline.

Contents

Select a chapter or page number to navigate. PDF bookmarks include every subsection.

1 Project overview and Release 1 scope	3
2 Functional requirements	3
3 Non-functional requirements	4
4 Release 1 architecture	4
5 MCP and RAG design	5
6 Validation and results	9
7 Integration summary	33
8 Known issues and limitations	33
9 Contributions, repository and showcase	34

1 Project overview and Release 1 scope

PropertyScope NSW connects property discovery, recorded sales, suburb context, site due diligence and buyer research. Release 1 retains five independently owned frontend/backend/database slices, their CRUD and AI-mode, and the shared research shell. It adds shared host MCP and RAG servers, grounded answers, and separate MCP/RAG loop validation modes. Cloud deployment and multi-agent orchestration belong to Release 2.

Matthew owns Shared and Feature 1; Burhan owns Feature 2; James owns Feature 3; Michael owns Feature 4; Derek owns Feature 5. Tutor approval for OpenAI and PostgreSQL was reconfirmed by Matthew on 3 October; this is a team statement, not an attached original approval document.

1.2 Delivery plan and risks

The shared runtime and Feature 1 landed first; the adoption kit then let feature owners register tools, corpora and backend routes independently. The final integration increment captures one running stack, checks each feature, refreshes the report and reviews the submission against the current rubric. The [release audit](#) records tasks, owners and remaining checks. Missing data, provider availability and scope changes are the main risks; versioned evidence, bounded failures and owner reviews limit their impact.

2 Functional requirements

ID	Required behaviour	Evidence
FR1	Retain each feature's UI, API and owned persistence	Live operations, Section 6.3
FR2	Discover registered MCP tools and return structured, scoped results	Tool catalogue and frontend captures
FR3	Retrieve only the selected feature's versioned guidance	Corpus probe and cited answers
FR4	Display citations, confidence and an insufficient-context response	Three assistant scenarios per feature
FR5	Retain Plan/Act/Observe/Adapt; add separate MCP/RAG validation modes	Captured outputs, Section 6.2
FR6	Deploy feature microservices in Compose; keep AI tier on host	Status and architecture evidence
FR7	Build/test each assigned workflow with MCP/RAG disabled	Five successful workflow links

3 Non-functional requirements

Quality	Target and validation boundary
Security	Token-protected host APIs; MCP/RAG bind to loopback. Live unauthenticated probes return 401. Signed calls bind feature, run, arguments, deadline and approval.
Isolation	No cross-feature imports or database access. Architecture checks enforce HTTP boundaries. Only Feature 1's two write tools require approval.
Grounding	Guidance findings cite retrieved passages; current facts cite successful tool calls. Corpus changes invalidate completion. Confidence describes evidence support, not probability.
Reliability	Timeouts and structured errors bound failed calls. RAG unavailability produces insufficient context; ordinary CRUD remains independent. Outage tests use controlled doubles.
Retrieval quality	Authored Feature 1 evaluation targets recall@5 ≥ 0.9 ; measured 0.963. This is not held-out five-feature answer accuracy.
Performance	Retrieval bounds top-k to five; registered tools have 2–10 second deadlines. The fresh RAG loop retrieves in 108 ms; one sample is not a latency benchmark.
Usability	Shared citations and activity remain keyboard accessible. Deterministic browser checks cover narrow layouts; live captures use Chromium at 1440×1000.
Reproducibility	Python 3.12, locked workspace dependencies, generated deployment manifests and versioned corpora. Tests require no provider credentials.
Maintainability	Corpus adoption changes only the owning manifest/documents; all five registrations pass generated-catalogue and architecture checks.
Interoperability	MCP Streamable HTTP and JSON Schema contracts; SDK round-trip tests require correlated, schema-valid results and reject unauthorised calls.
Availability	CRUD remains independent of unavailable AI; outage tests verify bounded degradation. Store-reopen and identical-ingestion tests require preserved history/version. These are recovery checks, not an uptime SLA.

4 Release 1 architecture

Compose runs the edge and five feature slices. Each database has one owning service. AI-mode contains the loop and calls the configured model over HTTPS; MCP and RAG are separate host processes. Backends reach AI-mode through `host.docker.internal`; MCP dispatches only to registered feature HTTP routes. Neither shared service opens a feature database.

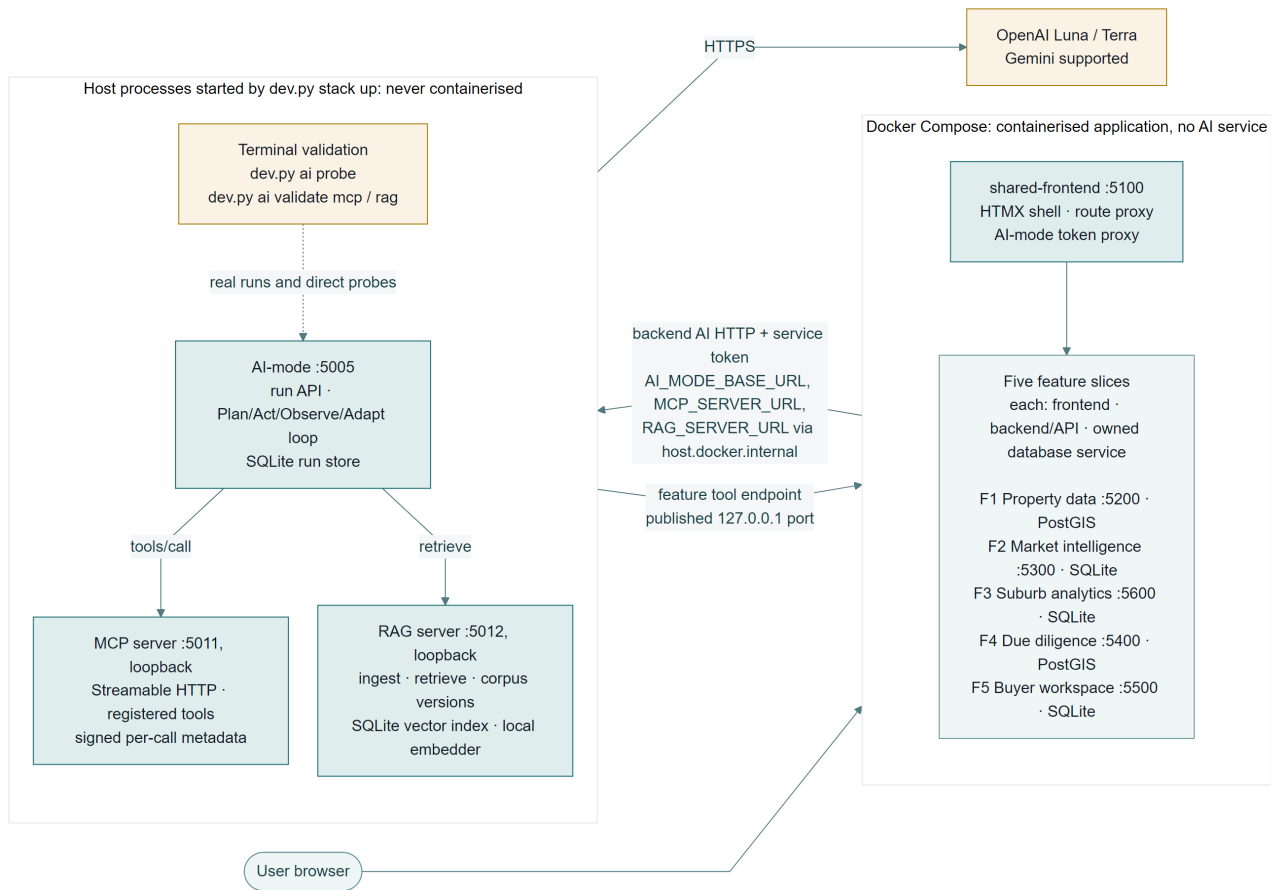


Figure 1 Overall Release 1 architecture and container boundary

Path	Responsibility
student-1/ ... student-5/	Owned UI, API, persistence, tools and corpus
ai-services/agent-core/	Loop policies, grounding and confidence
ai-services/ai-mode/	Run API, provider and MCP/RAG adapters
ai-services/mcp-server/, rag-server/	Tool transport; local semantic retrieval
shared/contracts/, tool-runtime/, frontend/	Schemas, dispatch and research shell
deployment/, .github/workflows/, scripts/	Compose projections, CI and developer commands

5 MCP and RAG design

A question goes from the frontend to its owning backend, then AI-mode with a fixed feature scope. The planner selects allowlisted tools. MCP validates signed invocation metadata, dispatches to the owning backend and validates correlation and output schemas. Successful MCP results carry `transport:mcp`; an ordinary HTTP result cannot satisfy the capture check.

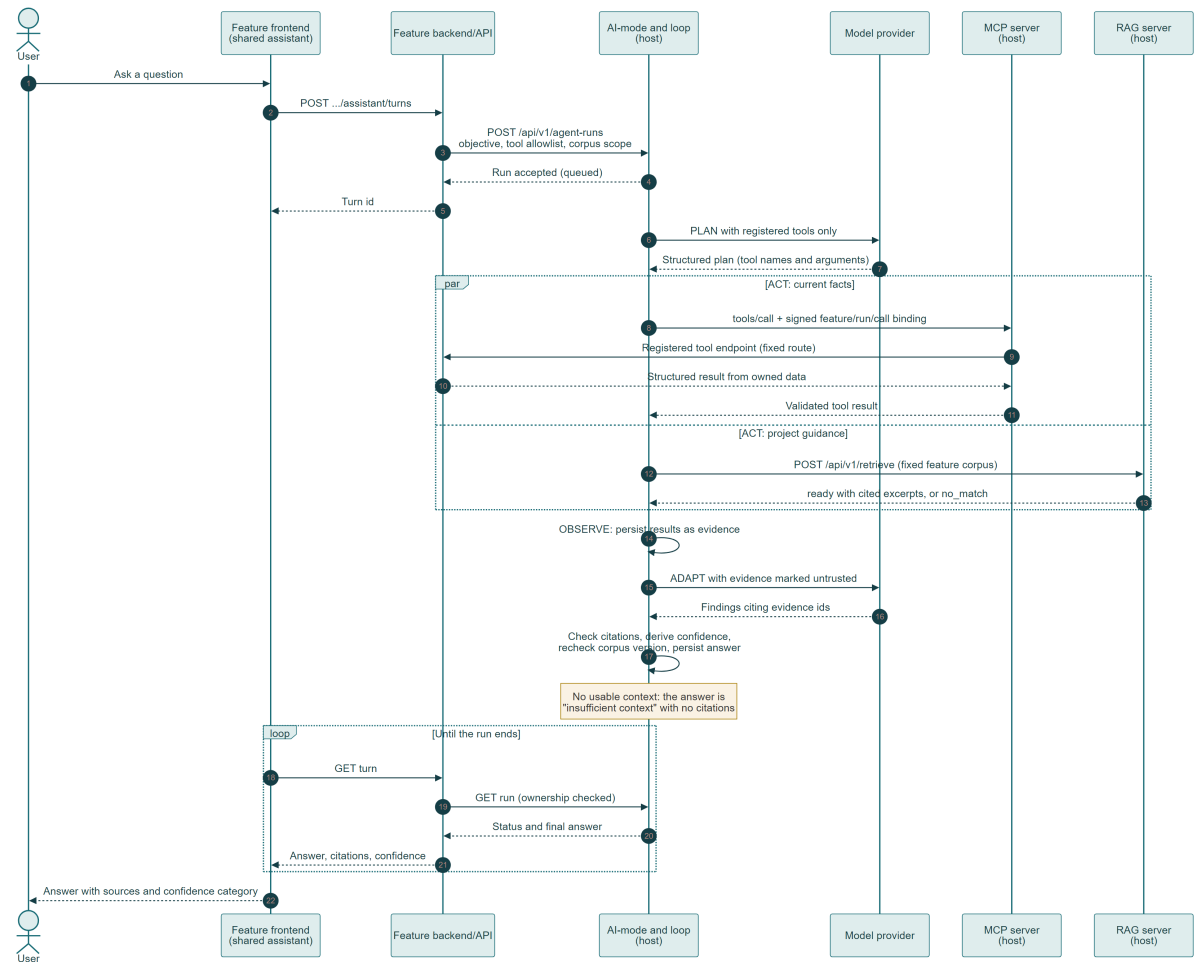


Figure 2 Frontend, backend, MCP and RAG interaction flow

5.1 Tools, inputs and outputs

MCP lists 29 feature tools. `context.retrieve.v1` is an internal AI-mode retrieval operation, not a thirtieth MCP tool. Full JSON schemas remain in each feature's `tool-catalog.yaml`.

Student	Feature	Registered MCP tools	Access	RAG corpus
1	Data Platform and Property Discovery	<code>platform.capabilities.v1</code> , <code>data.sources.v1</code> , <code>data.releases.v1</code> , <code>data.runs.v1</code> , <code>data.release_inspect.v1</code> , <code>data.run_inspect.v1</code> , <code>data.run_explain.v1</code> , <code>data.release_compare.v1</code> , <code>data.coverage.v1</code> , <code>property.search.v1</code> , <code>property.inspect.v1</code> , <code>property.locality_summary.v1</code> , <code>data.run_retry.v1</code> , <code>data.release_publish.v1</code>	12 read-only, 2 write, 2 need approval	operator-guidance (19 documents)
2	Property Sales Explorer and Market Cases	<code>market.capabilities.v1</code> , <code>market.cases.inspect.v1</code> , <code>market.sales.summary.v1</code>	3 read-only	operator-guidance (3 documents)
3	Suburb, Crime and Liveability Analytics	<code>suburb.published-context.v1</code> , <code>suburb.snapshot.v1</code> , <code>crime.compare.v1</code> , <code>crime.methodology.v1</code>	4 read-only	suburb-analytics-guidance (8 documents)

Student	Feature	Registered MCP tools	Access	RAG corpus
4	Site, Planning and Building Due Diligence	duediligence.capabilities.v1, duediligence.review.inspect.v1, duediligence.evidence.summary.v1	3 read-only	operator-guidance (3 documents)
5	Buyer Journey Workspace	buyer.capabilities.v1, buyer.cases.inspect.v1, buyer.notes.list.v1, buyer.tasks.list.v1, buyer.evidence.collect.v1	5 read-only	operator-guidance (5 documents)
Shared	Grounded runs	context.retrieve.v1, added by AI-mode to runs of features with a registered corpus	Read-only	The feature's own corpus

Tool	Required inputs	Output fields	Side effect	Timeout
platform.capabilities.v1	None	revision, application, assistant, features, suggested_questions, feature_status_meaning	read-only	3000 ms
data.sources.v1	None	items, count	read-only	3000 ms
data.releases.v1	None	items, count	read-only	3000 ms
data.runs.v1	None	items, count	read-only	3000 ms
data.release_inspect.v1	release_id	release, quality_results, quality_summary, receipts, activations, consumer_imports, publication_policy, accepted_predecessor, release_contract	read-only	3000 ms
data.run_inspect.v1	run_id	run, tasks, quality_results	read-only	3000 ms
data.run_explain.v1	run_id	run, tasks, quality_results, insight	read-only	3000 ms
data.release_compare.v1	candidate_release_id, predecessor_release_id	candidate, predecessor, differences	read-only	3000 ms
data.coverage.v1	locality	items	read-only	3000 ms
property.search.v1	query	items, count	read-only	3000 ms
property.inspect.v1	property_ref	property, identifiers, aliases, coverage, sales_history	read-only	3000 ms
property.locality_summary.v1	None	availability, scope, total_registered_addresses, unit_number_summary, bounding_box, top_streets, release	read-only	5000 ms
data.run_retry.v1	run_id, idempotency_key	child_run_id, replayed	reversible-write, approval	5000 ms
data.release_publish.v1	release_id, idempotency_key	status, receipt_id, replayed	destructive-write, approval	5000 ms

Tool	Required inputs	Output fields	Side effect	Timeout
market.capabilities.v1	None	revision, feature, tools, limitations, suggested_questions	read-only	3000 ms
market.cases.inspect.v1	market_case_id	market_case	read-only	3000 ms
market.sales.summary.v1	market_case_id	summary, sales	read-only	3000 ms
suburb.published-context.v1	locality	locality, status, population, schools, crime_coverage, sources, limitations	read-only	4000 ms
suburb.snapshot.v1	locality	suburb, places, limitations	read-only	3000 ms
crime.compare.v1	localities, from_month, to_month, measure	series, measure, offence, source_release, limitations	read-only	4000 ms
crime.methodology.v1	None	geography, measures, offence_categories, zero_missing_rule, rate_method, responsible_use, source_release	read-only	2000 ms
duediligence.capabilities.v1	None	revision, feature, tools, limitations, suggested_questions	read-only	3000 ms
duediligence.review.inspect.v1	site_review_id	site_review	read-only	3000 ms
duediligence.evidence.summary.v1	site_review_id	constraints, buildings	read-only	3000 ms
buyer.capabilities.v1	None	feature_key, purpose, limitations, suggested_questions, read_only	read-only	5000 ms
buyer.cases.inspect.v1	buyer_case_id	buyer_case, shortlisted_properties, property_count, limit, content_is_untrusted	read-only	5000 ms
buyer.notes.list.v1	buyer_case_id	items, count, limit, content_is_untrusted	read-only	5000 ms
buyer.tasks.list.v1	buyer_case_id	items, count, limit, content_is_untrusted	read-only	5000 ms
buyer.evidence.collect.v1	buyer_case_id	state, sections, evidence_references, limitations, bounds, content_is_untrusted	read-only	10000 ms

5.2 Knowledge sources and grounded responses

Each corpus contains authored CC0 project guidance, not current property facts. Sources retain document identifiers, dates and immutable corpus versions. The manifests link every Markdown source: [F1](#), [F2](#), [F3](#), [F4](#), [F5](#).

F1 covers publication and property-data limits; F2 explains sale matching and case eligibility; F3 distinguishes missing locality evidence from zero; F4 explains review evidence states; F5 explains case stages, shortlists and tasks. Live versions contain 19/3/8/3/5 documents.

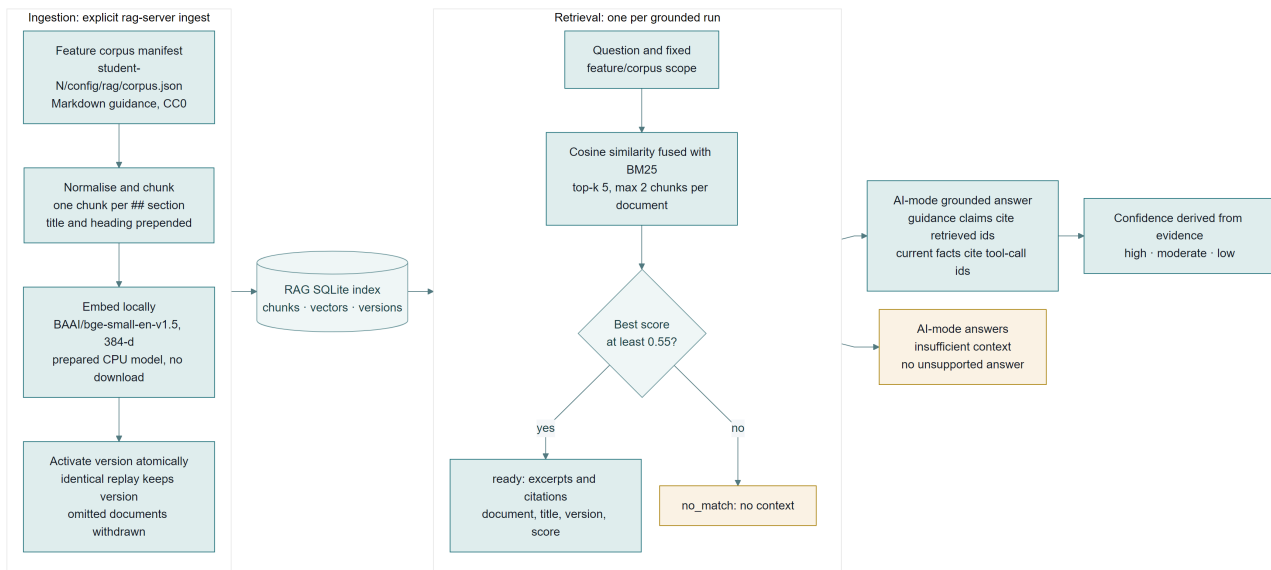


Figure 3 Versioned ingestion, retrieval and grounding

The prepared BGE model embeds passages locally. Retrieval combines semantic and lexical ranking, limits repeated documents and applies a 0.55 relevance floor. Retrieved text is untrusted evidence. Eligible grounded production runs use default.v9; ungrounded runs retain default.v7. High confidence needs strong support and no reported gaps; one document plus one current tool check can count as two supports. Reported gaps or weak support give moderate; model-reported stale/conflicting evidence can retain low. An unsupported answer has no citations.

6 Validation and results

6.1 Terminal validation

On 3 October, the [live probe](#) passed authentication, all five MCP registrations, corpus readiness, relevant retrieval and off-topic no-match checks. AI-mode reports healthy provider/store readiness and accepts its token; this alone does not prove a model answer. Feature screenshots below separately exercise the configured provider.

```

$ uv run scripts/dev.py ai probe
MCP: HTTP 200; 29 tools; feature counts 14 / 3 / 4 / 3 / 5
RAG: BAAI/bge-small-en-v1.5 ready; 384 dimensions
Corpora: 19 / 3 / 8 / 3 / 5 documents
Each feature: relevant retrieval PASS; off-topic no_match PASS
Unauthenticated AI-mode / MCP / RAG: HTTP 401
All checks passed.
  
```

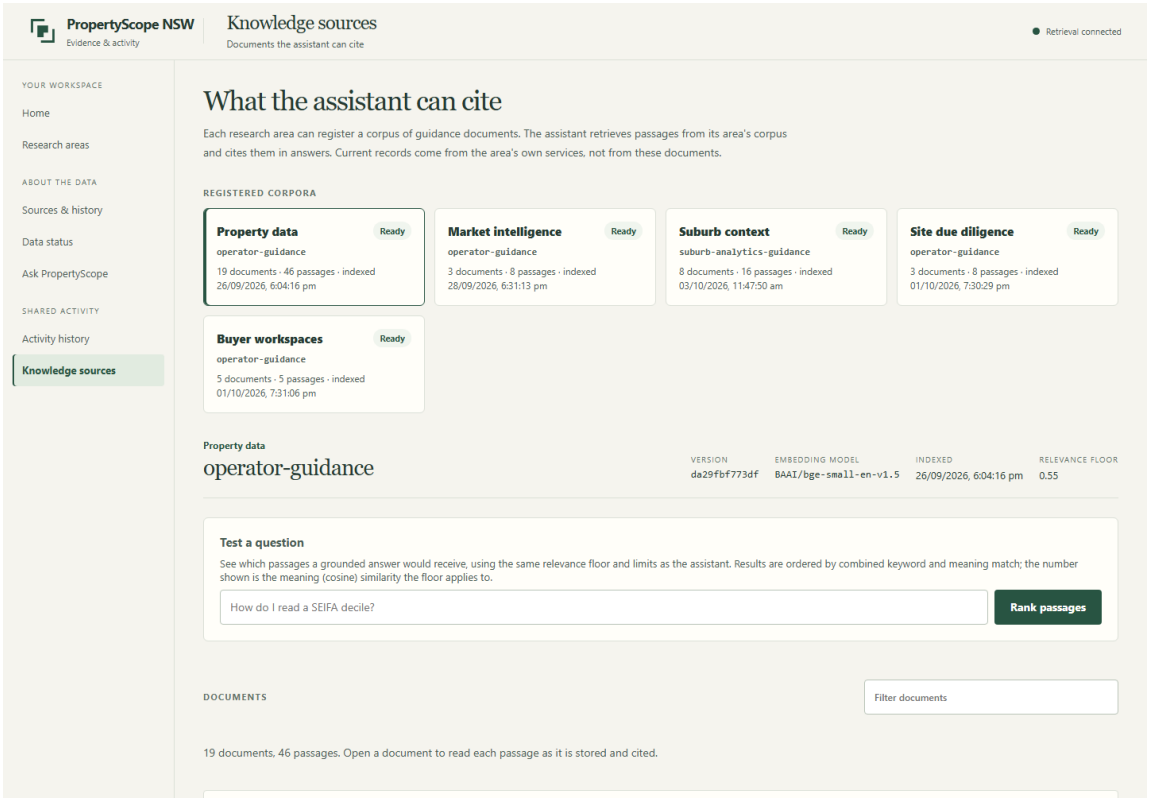


Figure 4 Registered live knowledge sources

6.2 Shared agentic loop

Both named validation modes run the production loop against live local services with deterministic default v8 decisions. They prove transport and phase execution; they do not measure provider answer quality. The ordinary production mode remains available.

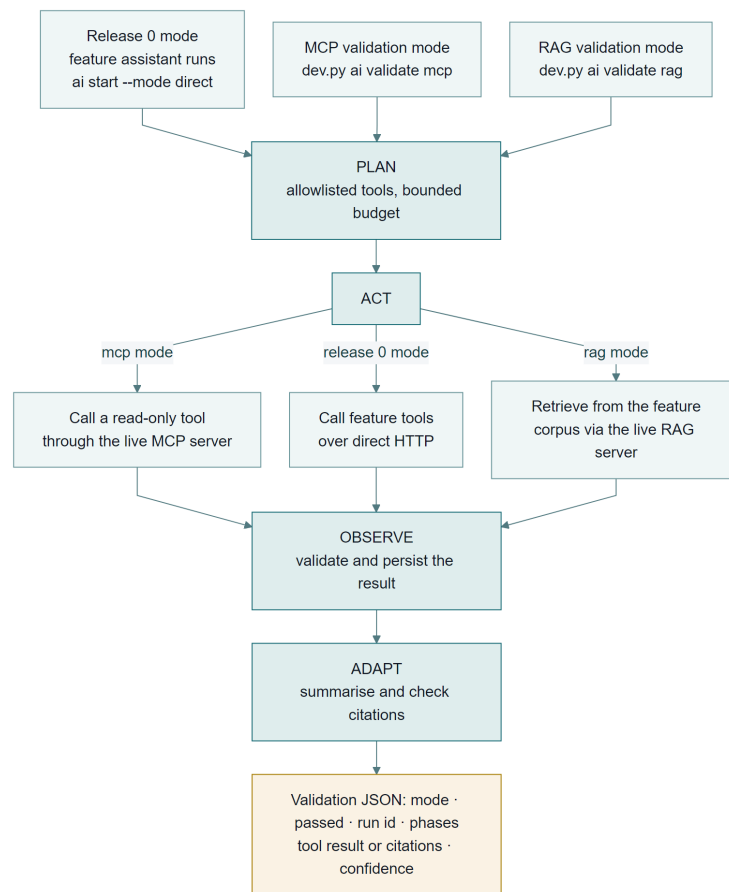


Figure 5 Loop entry points and validation modes

```

$ uv run scripts/dev.py ai validate mcp
mode      mcp  passed=true  status=succeeded
feature   student-1-property-scope-data-platform
tool      platform.capabilities.v1
corpus    -
run_id    97ea140e-2a1c-4f23-a1bf-c1a4f6715432
request_id r1-mcp-654a814d-bab5-4b34-9d09-7eac0adf450d
phases    plan -> act -> observe -> adapt
transport live-local-services  decisions=deterministic-validation
tool call  f5b53440-94a7-4730-b3a2-d8b397e969e0: succeeded in 15 ms; fields: application,
assistant, feature_status_meaning, features
summary    Registered MCP tool returned structured evidence.
boundary   Production runner and local services; deterministic model decisions, not provider
answer quality.

$ uv run scripts/dev.py ai validate rag
mode      rag  passed=true  status=succeeded
feature   student-1-property-scope-data-platform
tool      context.retrieve.v1
corpus    operator-guidance
run_id    3e319a2e-1667-4f36-9397-2f7502d8fd34
request_id r1-rag-073dd32b-de85-44ec-a822-ba01bb09a474
phases    plan -> act -> observe -> adapt
transport live-local-services  decisions=deterministic-validation
tool call  0516ef7b-bbf9-474b-a646-2ebbb9522c63: succeeded in 108 ms, retrieval ready, 5 citations
version    da29fbf773df88ce8a0939cae9f8ade1ca094441e368ac3fba2ac6c9c1cb1fed
cites      publication: Publishing a version
cites      glossary: Property data terms
cites      data-updates: Running a data update
final      confidence=moderate  grounding=ready
summary    Retrieved project guidance.
boundary   Production runner and local services; deterministic model decisions, not provider
answer quality.
  
```

6.3 Feature UI, API and persistence

The [live operations record](#) identifies frontend responses, backend reads and database-backed operations for each feature. Assistant captures use existing saved contexts. Viewport captures use 1440×1000; focused cards reflow at 1100 pixels wide. Each capture sidecar records the source SHA, time, run/request IDs, successful tool calls, corpus version and image hashes; no private reasoning trace is exported.

Feature	Live persistence observation
1	Draft source create/read/update/delete; cleanup GET 404
2	Owning store: 10 cases, 26 sale records
3	Owning store: 10 suburbs, 400 indicators, 38 amenities, 10 overviews, 10 user-suburbs
4	Site review create/read/update/delete; cleanup GET 404
5	Owning store: 10 cases, 11 notes, 12 properties, 12 tasks

Counts include demonstration records; they establish persisted operations, not official coverage.

Feature 1 Property data

Full UI captures: [mcp](#), [rag](#), [insufficient](#). [Run metadata](#).

Published-release counts come from MCP; guidance explains why missing sale history is unknown.

PROPERTYSCOPE ASSISTANT

Complete

Six datasets have accepted releases in the returned inventory, with record counts ranging from 3 to 7,402,643. The list below reports the accepted releases, not candidates or superseded releases.

Confidence: High

2 guidance sources · 1 record check

Based on 2 cited guidance documents and 1 current record check, each closely matching the question, with no evidence gaps.

Key findings

Accepted releases returned: nsw-psi-sales — 7,402,643 records; bocsar-crime — 318,122; gnaf-nsw — 5,190,134; abs-seifa-2021 — 4,320; fixture-property — 3; nsw-government-schools — 10.

Current record

Inspect source: Dataset releases

Publishing makes a reviewed candidate the current version of its dataset; candidates must not be presented as published data.

Guidance

Inspect source: Publishing a version

Inspect source: Where a value came from

Suggested next step

Sources & activity

Run b0b2b2b0... 3 Oct 2026, 12:25 pm

Open full activity

Figure 6 Feature 1 MCP result

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12

PROPERTYSCOPE ASSISTANT

● Complete

A property can show no recorded sales because PropertyScope only displays published sale records that match exactly one G-NAF address. An empty sale history is missing matched evidence, not proof that the property has never sold.

● Confidence: Moderate

2 guidance sources

Based on 2 cited guidance documents; the answer notes 1 evidence gap.

Key findings

Sales with ambiguous or unrecognised addresses are retained as unmatched rather than linked by guesswork. Older free-text records and unit sales are more likely to remain unmatched; a sale may also be older than 1990 or not yet published.

Guidance

Inspect source: NSW property sales (PSI)

“No published sale history” means no matched sale was found in the published NSW sales data; it does not establish that no sale occurred.

Guidance

Inspect source: Reading a property record

Evidence gaps

No exact property reference was supplied, so the cause for a particular property's missing sale history cannot be established.

► Suggested next step

Sources & activity

Run 421e03a2... 3 Oct 2026, 12:26 pm [Open full activity](#)

Figure 7 Feature 1 cited answer and confidence

▼ NSW property sales (PSI)

● Project guidance

Project guidance explains the workflow; it does not establish current property conditions.

How sales are matched to a property

A sale is linked to a property record only when its address resolves to exactly one published G-NAF address. The match is shown with its method and confidence on the Sale history tab. If the address is ambiguous or not found, the sale is kept but left unmatched rather than guessed.

Matching is incomplete. In the September 2026 load, about 69% of post-2001 sale records and 54% of pre-2001 records matched a single address. Unit sales and older records written in free-text address formats are the most likely to stay unmatched.

An empty Sale history tab means no matched sale was found in the published sales data. It does not mean the property has never sold: the sale may be unmatched, older than 1990, or not yet published.

Location
Property data guidance: sales-history; section How sales are matched to a property; chars 785-1554

Source date
2026-09-26

Indexed
26 Sept 2026, 6:04 pm

Corpus
operator-guidance

Version
da29fbf773df88ce8a0939cae9f8ade1ca094441e368ac3fba2ac6c9c1cb1fed

Document / chunk
sales-history / 314ac51b545f054de3370a284397d237513cd50947e6323c3e0f1a0217c80018

Citation ID
cite-f4c9c4313f6a963f1897eace96cc3f316c1bcaa0a501cf9194e4052c416391b7

Content hash
a2a23bd56b60ca531cd994565d677166838979584ec34c474d74106bff31d7ef

[Open source document \(new tab\)](#)

Feature 1 supporting passage, source date and corpus version

PROPERTYSCOPE ASSISTANT

Complete

I can't advise on oven temperature for a chocolate cake because this assistant only supports PropertyScope property-data research. Check the specific cake recipe's baking instructions.

Confidence: Insufficient

1 record check

Relevant document context is unavailable or insufficient.

Key findings

This assistant can inspect only Property data and its available capabilities concern NSW property research, rather than cooking guidance.

Current record

Inspect source: Available research tools

Evidence gaps

Similarity measures retrieval relevance, not factual confidence

No grounded cooking or recipe guidance was retrieved for chocolate-cake oven temperatures.

Suggested next step

Sources & activity

Run 78ca25c8... 3 Oct 2026, 12:26 pm [Open full activity](#)

Figure 8 Feature 1 insufficient context

Feature 2 Market cases

Full UI captures: [mcp](#), [rag](#), [insufficient](#). [Run metadata](#).

The selected case constrains its eligible sales and recorded-price summary.

PROPERTYSCOPE ASSISTANT

• Complete

Sydney example – full history for 11 Example Street, Sydney NSW 2000 has 0 eligible sales. No median recorded price is available because there are no eligible sale records.

• Confidence: High

2 record checks

Based on 2 current record checks, each closely matching the question, with no evidence gaps.

Key findings

The case has 0 eligible sales, 0 excluded sales, no transaction-volume entries and no source releases for the 2019–2026 period at the selected tier B threshold. The median recorded price is unavailable.

Current recordInspect source: Market sales summary

The inspected case is Sydney example – full history for 11 Example Street, Sydney NSW 2000.

Current recordInspect source: Market cases inspect

With fewer than three eligible sales, the evidence is insufficient; recorded sales are historical evidence and not a valuation or buying recommendation.

Current recordInspect source: Market sales summary

► Suggested next step

Sources & activity

Runa31a499d...3 Oct 2026, 12:26 pm

Open full activity

Figure 9 Feature 2 MCP result

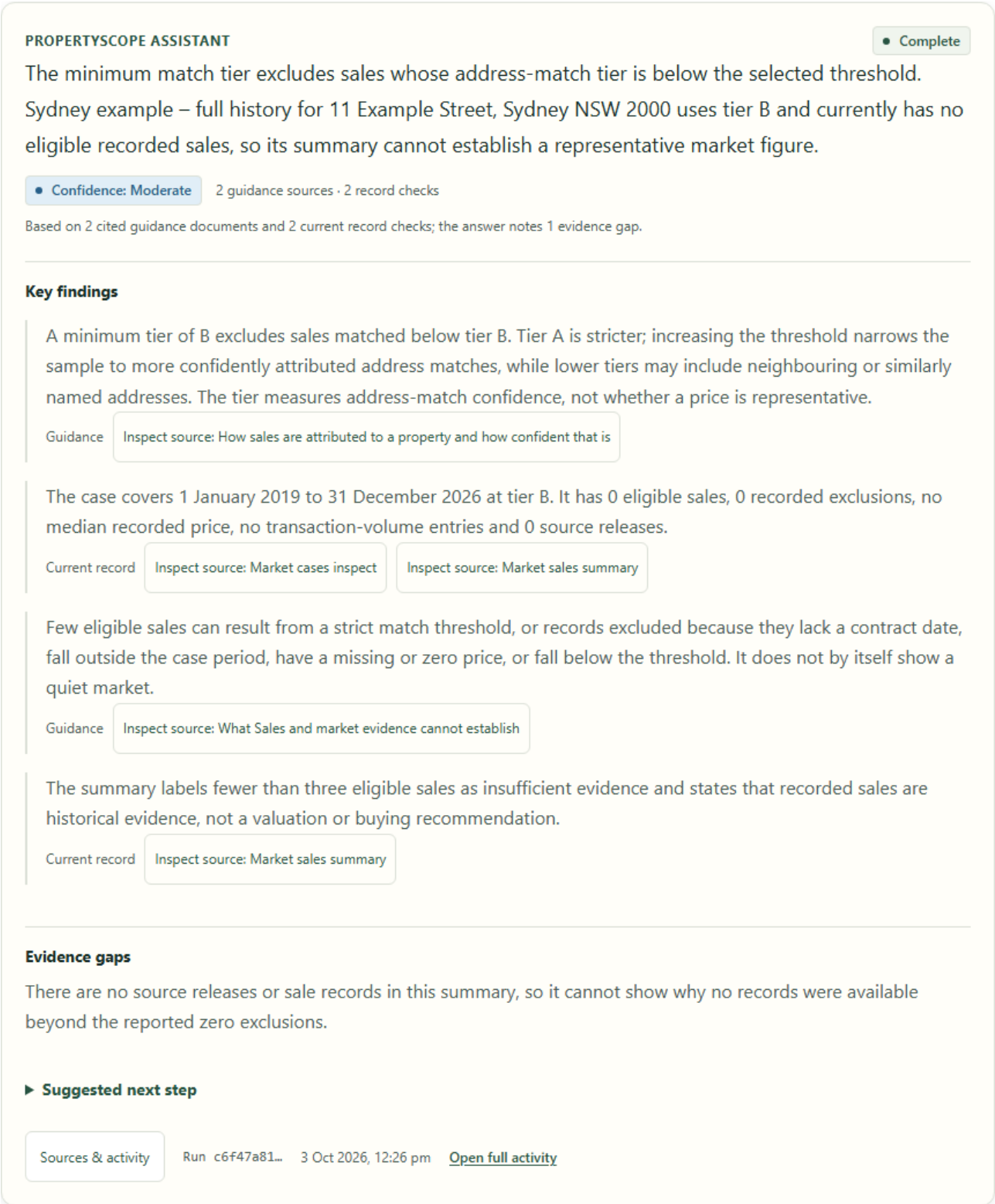


Figure 10 Feature 2 cited answer and confidence

▼ What Sales and market evidence cannot establish

● Project guidance

Project guidance explains the workflow; it does not establish current property conditions.

Coverage gaps and excluded records

A summary reports only the sales that passed the case filters, and separately counts what it left out. Records are excluded for four reasons: no contract date, a contract date outside the case period, a missing or zero price, or a match tier below the threshold you selected.

Excluded records are not errors and are not deleted. Raising the minimum match tier deliberately narrows the evidence, so a case with few eligible sales may reflect a strict threshold rather than a quiet market.

Where fewer than three sales are eligible, the summary states that this is insufficient evidence. Treat the median from such a case as an observation about those specific records, not a representative figure.

Location
Operator guidance: evidence-limits; section Coverage gaps and excluded records; chars 1169-1905

Source date
2026-09-28

Indexed
28 Sept 2026, 6:31 pm

Corpus
operator-guidance

Version
fc79eb7f58187a25419ecc32b714dfb061d7e9c4e3233358bad4c98101b881d2

Document / chunk
evidence-limits / ecd74b4739d4def521295212538ff53ccf38197be842e39ec94e3622e67cd369

Citation ID
cite-c70c0ee60171876d19c9f70e65ddc7be023393abc58337a9b19395058faac697

Content hash
22a036de43173fa7388afdd5b2072a4844c459d69209a081d7439d7be7e78d29

[Open source document \(new tab\)](#)

Feature 2 supporting passage, source date and corpus version

PROPERTYSCOPE ASSISTANT

• Complete

Insufficient context to provide a grounded explanation.

• Confidence: Insufficient

Relevant document context is unavailable or insufficient.

Evidence gaps

Similarity measures retrieval relevance, not factual confidence

Password-reset instructions or an authorised account-support capability were not available.

► Suggested next step

Sources & activity

Run 0dbebe76... 3 Oct 2026, 12:26 pm [Open full activity](#)

Figure 11 Feature 2 insufficient context

Feature 3 Suburb analytics

Full UI captures: [mcp](#), [rag](#), [insufficient](#). [Run metadata](#).

The MCP example returns crime methodology; the cited answer distinguishes missing evidence from zero.

PROPERTYSCOPE ASSISTANT

● Complete

A recorded zero is 0 only within the source’s declared observation universe; missing or unavailable evidence is null and must not be treated or displayed as zero. This feature provides count and rate measures; its fixture rate is count divided by fixture population, multiplied by 100,000.

● Confidence: Moderate

2 guidance sources · 1 record check

Based on 2 cited guidance documents and 1 current record check; the answer notes 1 evidence gap.

Key findings

The feature’s methodology defines a recorded zero as 0 and missing evidence as null. Missing or unavailable evidence must not be plotted or described as zero, and sparse coverage does not justify assumptions about unobserved months.

Guidance

Inspect source: How to interpret recorded crime evidence

Inspect source: Published evidence and demonstration data

The available measures are count and rate. The configured fixture calculation is: fixture count ÷ fixture population × 100,000; the source release is demo-2026.1.

Current record

Inspect source: Crime methodology

A count is recorded observations for the selected categories and period, while a rate uses a stated population denominator. They are not interchangeable, and comparisons must not mix them; differing count and population reference periods should be disclosed.

Guidance

Inspect source: How to interpret recorded crime evidence

The methodology response describes an NSW suburb fixture, so its rate method and release identify demonstration data rather than current NSW conditions.

Guidance

Inspect source: Published evidence and demonstration data

Evidence gaps

No published locality-specific population denominator, observation universe or reference period was provided for calculating a current locality rate.

► Suggested next step

Sources & activity

Run 820c8083... 3 Oct 2026, 12:27 pm [Open full activity](#)

Figure 12 Feature 3 MCP result

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20

PROPERTYSCOPE ASSISTANT

● Complete

Feature 3 displays a recorded zero as 0 only where the source observed the relevant category and period. Missing or unavailable evidence remains null, not zero.

● Confidence: High

2 guidance sources · 1 record check

Based on 2 cited guidance documents and 1 current record check, each closely matching the question, with no evidence gaps.

Key findings

The Feature 3 methodology rule is: recorded zero is 0; missing evidence is null. The methodology source release is demo-2026.1 and applies to an NSW suburb fixture.

Current record

Inspect source: Crime methodology

A zero is meaningful only within the source’s declared observation universe. Missing or unavailable evidence must not be plotted or described as zero, and sparse coverage does not justify assumptions about unobserved months.

Guidance

Inspect source: How to interpret recorded crime evidence

Unavailable evidence is not evidence that an event or other characteristic does not exist.

Guidance

Inspect source: Published evidence and demonstration data

► Suggested next step

Sources & activity

Run cc21e6f6... 3 Oct 2026, 12:27 pm [Open full activity](#)

Figure 13 Feature 3 cited answer and confidence

▼ How to interpret recorded crime evidence

• Project guidance

Project guidance explains the workflow; it does not establish current property conditions.

Zero and missing

A recorded zero is meaningful only inside the source's declared observation universe. Missing or unavailable evidence remains null and must not be plotted or described as zero. Sparse coverage does not support an assumption about unobserved months.

Responsible interpretation

Recorded crime data does not establish why events occurred, an individual's risk, the safety of a place or future crime. A comparison is descriptive evidence for a controlled period and category, not a suburb ranking.

Location

Suburb analytics guidance: crime methodology; section Zero and missing; chars 412-930

Source date

2026-10-02

Indexed

3 Oct 2026, 11:47 am

Corpus

suburb-analytics-guidance

Version

8db8f59a9c9daa8ffd5126f157fa0097a1a83bf9208a2a6c9d81f41321ba3b4a

Document / chunk

crime-methodology / 454002b8ce681cb06eab0023edd13ac757dc8781f0763788a22d448a2bb5c24d

Citation ID

cite-1ba53d6262ea64ce9b5a200b9d02bedec4231df4e225bc7e8da2cf6b21e65a47

Content hash

f6c34c70ee3990754b221ff9fd1f6aba2f3d116d608e995846ed31924409efde

[Open source document \(new tab\)](#)

Feature 3 supporting passage, source date and corpus version

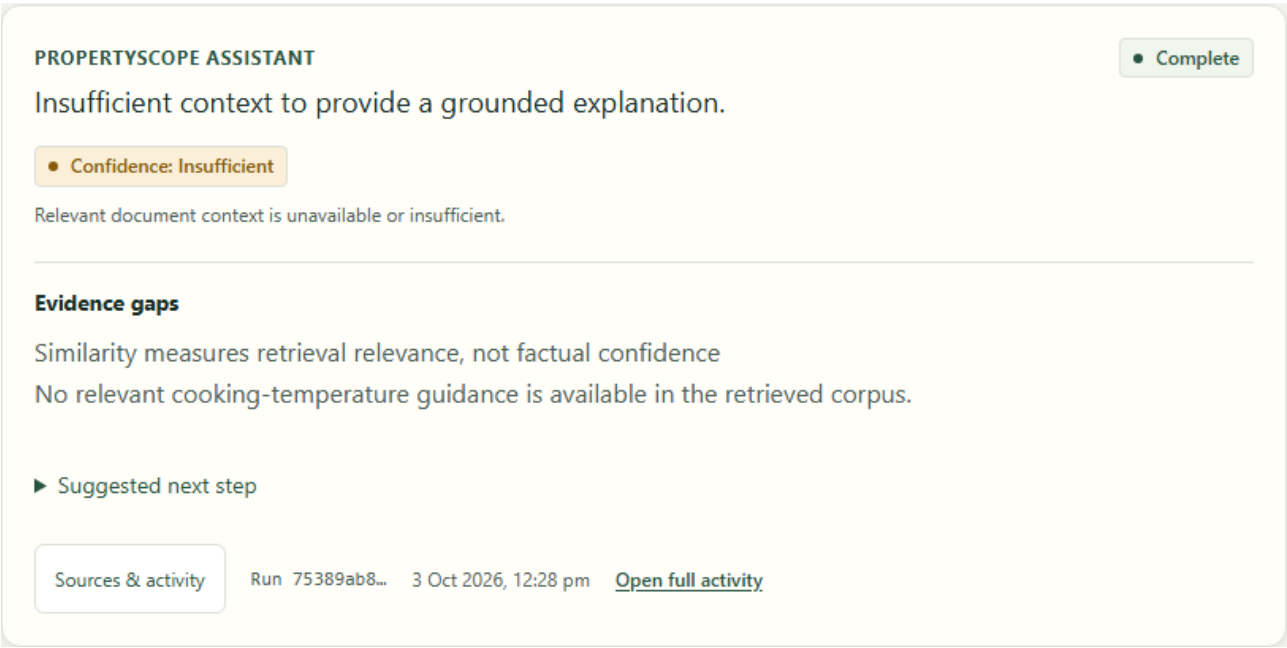


Figure 14 Feature 3 insufficient context

Feature 4 Due diligence

Full UI captures: [mcp](#), [rag](#), [insufficient](#). [Run metadata](#).

The selected review exposes evidence states without treating an absent layer as clearance.

PROPERTYSCOPE ASSISTANT

Complete

For 11 Example Street, Sydney NSW 2000, the review records planning controls, environmental map checks, strata information and building-related records in several evidence states. Some records are confirmed, while others have partial coverage, no mapped intersection or are unavailable.

Confidence: Moderate

2 guidance sources · 1 record check

Based on 2 cited guidance documents and 1 current record check; the answer notes 1 evidence gap.

Key findings

Planning evidence records confirmed R2 Low Density Residential zoning and a maximum floor-space ratio of 0.5:1. A maximum building height of 9.5 m has partial coverage, and a separate zoning map record is marked non-intersection.

Current record

Inspect source: Duediligence evidence summary

Environmental evidence records no intersection with the 1% AEP flood planning area. Bushfire-prone-land mapping (Vegetation Category 1) and the heritage listing record are unavailable, so the review cannot conclude either matter.

Current record

Inspect source: Duediligence evidence summary

Strata and building evidence includes a confirmed registered strata scheme with an active managing agent and a confirmed building-order or rectification-notice record. A further strata record is unavailable; tribunal matters are marked non-intersection; and written undertakings have partial coverage.

Current record

Inspect source: Duediligence evidence summary

Confirmed means a source record applies to the property; partial coverage means only part of the expected matter is covered; non-intersection means a mapped layer did not intersect the property; and unavailable means no conclusion can be drawn either way.

Guidance

Inspect source: How due-diligence evidence is classified and how confident that is

This review records gathered evidence only. It does not certify planning or Building Code compliance, safety, structural condition, approval history or legal suitability.

Guidance

Inspect source: What Site and due diligence evidence cannot establish

Evidence gaps

The unavailable bushfire, heritage and strata records do not establish the underlying position for those matters.

Suggested next step

Sources & activity

Run 9fce0ba9... 3 Oct 2026, 12:35 pm [Open full activity](#)

Figure 15 Feature 4 MCP result

PROPERTYSCOPE ASSISTANT

Complete

In a site review, confirmed means a source has an applicable record; non-intersection means a checked mapped layer did not overlap the property; partial means only part of the expected coverage is present; and unavailable means no conclusion can be drawn from that source. For 11 Example Street, each state appears in the planning, environmental, strata or building evidence, but none establishes legal suitability, compliance or safety.

Confidence: High

2 guidance sources · 1 record check

Based on 2 cited guidance documents and 1 current record check, each closely matching the question, with no evidence gaps.

Key findings

Confirmed means the source holds a record that applies to the property. At 11 Example Street, the review has confirmed zoning and floor-space-ratio controls, plus confirmed strata and building-order records.

Guidance

Inspect source: How due-diligence evidence is classified and how confident that is

Non-intersection means a mapped constraint layer was checked and did not intersect the property. The review records this state for the flood planning area and a zoning-control entry; it reduces concern for that specific layer only.

Guidance

Inspect source: How due-diligence evidence is classified and how confident that is

Inspect source: How due-diligence evidence is classified and how confident that is

Partial coverage means the source applies only partly to the property or addresses some, but not all, expected matters. The review marks building-height controls and written undertakings as partial coverage.

Guidance

Inspect source: How due-diligence evidence is classified and how confident that is

Unavailable means the source could not be reached or holds no record, so it is not evidence that a constraint is absent. The review has unavailable bushfire, heritage and strata-status evidence.

Guidance

Inspect source: How due-diligence evidence is classified and how confident that is

Inspect source: What Site and due diligence evidence cannot establish

The review evidence contains confirmed, non-intersection, partial-coverage and unavailable entries across planning, environmental, strata and building matters for 11 Example Street.

Current record

Inspect source: Duediligence evidence summary

Suggested next step

Sources & activity

Run 68f55947... 3 Oct 2026, 12:35 pm [Open full activity](#)

Figure 16 Feature 4 cited answer and confidence

▼ How due-diligence evidence is classified and how confident that is

● Project guidance

Project guidance explains the workflow; it does not establish current property conditions.

How due-diligence evidence is classified

Every planning, environmental, strata and building record in a review carries an evidence state and a confidence, so a reader can see not just what was found but how firmly it is established. The states used are confirmed, partial coverage, non-intersection and unavailable.

What each evidence state means

Confirmed means the source holds a record that applies to this property, such as a control that covers the land or a building record on file. Partial coverage means the source applies to the property only in part, or covers some of the matters you would expect but not all of them.

Non-intersection means a mapped constraint layer, such as a flood or bushfire category, was checked against the property and did not intersect it. Unavailable means the source could not be reached or holds no record, so nothing can be concluded either way. Confidence describes how sure the source is about its own record; it is not a statement of legal or structural certainty.

Location
Operator guidance: evidence-states; section How due-diligence evidence is classified; chars 1-1016

Source date
2026-10-01

Indexed
1 Oct 2026, 7:30 pm

Corpus
operator-guidance

Version
b50761eebb33c163daecab4e0ab7fc4dbf662c69a64728d509f740479036c60e

Document / chunk
evidence-states / c4e8665526d09b2dc445bd1f879670eddd2329a65c1c5b8df06600ce6866770c

Citation ID
cite-670b34f073e817dd67b87c675ae7240f35cc43ff60e7f6c2393de999a5882e5b

Content hash
78e1d7f2add1676ca291128d18a1f0e01900dc77f3fa6738f850d3babbed0851

[Open source document \(new tab\)](#)

Feature 4 supporting passage, source date and corpus version

PROPERTYSCOPE ASSISTANT

Complete

I can't establish how to reset your password from the available guidance. This feature supports site due-diligence research only, not account management.

Confidence: Insufficient

1 record check

Relevant document context is unavailable or insufficient.

Key findings

The available feature provides read-only site-review and property-evidence functions; it has no password-reset capability.

Current record

Inspect source: Duediligence capabilities

Evidence gaps

Similarity measures retrieval relevance, not factual confidence

No password-reset instructions or account-support guidance were available.

Suggested next step

Sources & activity

Run 45b06a7c... 3 Oct 2026, 12:36 pm [Open full activity](#)

Figure 17 Feature 4 insufficient context

Feature 5 Buyer workspace

Full UI captures: [mcp](#), [rag](#), [insufficient](#). [Run metadata](#).

The selected buyer case supplies its budget, target suburbs and shortlist.

PROPERTYSCOPE ASSISTANT

Complete

The recorded budget is \$850,000–\$1,250,000, with SYDNEY, NSW as the target suburb. The shortlist contains 11 Example Street, Sydney NSW 2000 and 2 Sample Avenue, Mosman NSW 2088.

Confidence: Moderate

1 guidance source · 2 record checks

Based on 1 cited guidance document and 2 current record checks; the answer notes 1 evidence gap.

Key findings

The active case records an apartment preference, priorities of transport and planning evidence, a budget range of \$850,000 to \$1,250,000, and the target suburb SYDNEY, NSW.

Current record

Inspect source: Buyer cases inspect

The validated shortlist contains 11 Example Street, Sydney NSW 2000 (Reviewing; high priority; rating 4) and 2 Sample Avenue, Mosman NSW 2088 (Shortlisted; medium priority; rating 3).

Current record

Inspect source: Buyer cases inspect

Two tasks remain incomplete: confirm unassessed priorities by 15 September 2026, and verify due-diligence coverage with a qualified professional for 11 Example Street by 12 September 2026.

Current record

Inspect source: Buyer tasks list

Shortlist stages organise the buyer workflow, and ratings and priorities are personal assessments rather than valuations.

Guidance

Inspect source: Shortlists and journey stages

Evidence gaps

Property discovery evidence needs verification, Sales research evidence is conflicting, Suburb analytics evidence is unavailable, and Due diligence evidence is partial. The target suburb is a buyer preference, not proof of a property location or suitability.

Suggested next step

Sources & activity

Run 294a29c7... 3 Oct 2026, 12:29 pm [Open full activity](#)

Figure 18 Feature 5 MCP result

PropertyScope NSW | Release 1 Technical Report

28

PROPERTYSCOPE ASSISTANT

Complete

Journey stages organise your shortlist workflow: Shortlisted, Inspecting, Reviewing, Offer Considered and Closed. Your active Sydney apartment case has two validated properties: one Shortlisted and one Reviewing; use the stages, priorities and ratings to focus review work, not as value or suitability judgments.

Confidence: Moderate

2 guidance sources · 3 record checks

Based on 2 cited guidance documents and 3 current record checks; the answer notes 2 evidence gaps.

Key findings

Guidance

Inspect source: Shortlists and journey stages

The workspace stages are Shortlisted, Inspecting, Reviewing, Offer Considered and Closed. They organise workflow only: changing a stage does not arrange an inspection, submit an offer or buy a property. Ratings and priorities are personal assessments, not valuations.

Current record

Inspect source: Buyer cases inspect

The active case targets Sydney, NSW, seeks an apartment, and records transport and planning evidence as priorities. Its budget preference is AUD 850,000–1,250,000.

Current record

Inspect source: Buyer cases inspect

The shortlist contains two validated properties: 2 Sample Avenue, Mosman is Shortlisted with medium priority and a rating of 3; 11 Example Street, Sydney is Reviewing with high priority and a rating of 4.

Current record

Inspect source: Buyer tasks list

Two tasks remain incomplete: confirm unassessed priorities, and verify due-diligence coverage with a qualified professional for the Reviewing property. One sales-exclusions review task is complete.

Current record

Inspect source: Buyer evidence collect

Current cross-feature evidence is partial: Property discovery needs verification, Sales research is conflicting, Suburb analytics is unavailable, and Due diligence is partial.

Guidance

Inspect source: Buyer notes and tasks

Incomplete tasks remain visible work to review, and completing a task changes only the saved buyer record rather than carrying out external work.

Evidence gaps

Property discovery requires verification; Sales research contains conflicting information; Suburb analytics is unavailable; and Due diligence coverage is partial.

The collected evidence is bounded and does not establish valuation, affordability, condition or suitability.

► Suggested next step

Sources & activity

Run 3a3e52a1... 3 Oct 2026, 12:29 pm [Open full activity](#)

Figure 19 Feature 5 cited answer and confidence

▼ Shortlists and journey stages

• Project guidance

Project guidance explains the workflow; it does not establish current property conditions.

Shortlists and journey stages

Guidance

Add a property using its Property discovery reference. The workspace checks that reference through the public property API. Pending or unavailable validation does not establish a property identity. Review the address and source evidence before relying on it. Journey stages are Shortlisted, Inspecting, Reviewing, Offer Considered and Closed. They organise the buyer's workflow; changing a stage does not book an inspection, submit an offer or purchase a property. Ratings from one to five and low, medium or high priorities are personal assessments, not valuations.

Location

Buyer workspace guidance v1: shortlist-and-stages; chars 1-612

Source date

2026-10-01

Indexed

1 Oct 2026, 7:31 pm

Corpus

operator-guidance

Version

f35d5f4e808222012131b29f3e220cdacc03d823c127700a1224ea2663ed91bd

Document / chunk

shortlist-and-stages / 0ff0b395d60cc9be05a549eecef82a6c36b79d3771c17de97baeb84923c27d74

Citation ID

cite-3be027f428f29748c4104a8993443c0353d41555a997218b8aa8e0aea9af6994

Content hash

cef37a0b2527fcfb1e1fe7770f2b79c1aa2af1770b0aa7ab396335d0f07df50b

[Open source document \(new tab\)](#)

Feature 5 supporting passage, source date and corpus version

PROPERTYSCOPE ASSISTANT

Complete

Insufficient context to provide a grounded explanation.

Confidence: Insufficient

Relevant document context is unavailable or insufficient.

Evidence gaps

Similarity measures retrieval relevance, not factual confidence

A relevant recipe or cooking guidance source specifying the oven temperature, including whether the setting is fan-forced or conventional, is unavailable.

Suggested next step

Sources & activity

Run 4ae48dda... 3 Oct 2026, 12:29 pm [Open full activity](#)

Figure 20 Feature 5 insufficient context

6.4 Retrieval evaluation

The authored Feature 1 baseline uses a disposable index and the same embedding model. Expected-source recall measures document retrieval, not factual correctness of a generated answer.

Measure	Result
Corpus	19 documents, 46 chunks, version da29fbf773df
Embedding	BAAI/bge-small-en-v1.5, 384 dimensions, local CPU
Settings	top-k 5, relevance floor 0.55
Supported questions	27; expected-source recall@5 0.963
Questions the corpus does not cover	4 of 6 retrieve no context
Evaluated	2026-09-26

6.5 GitHub Actions

All assigned workflows set AI_MODE_MCP_ENABLED=false and AI_MODE_RAG_ENABLED=false and test contracts with doubles. The following successful runs cover integrated baseline 330e65f, 1 October UTC, including all five merged Release 1 implementations. They precede this report-update branch; [run metadata](#) preserves their full SHAs. Student 5 starts offline direct AI-mode; MCP/RAG remain disabled. Its owner should reconcile this with the brief’s broader AI-mode-disabled instruction.

Workflow	Successful run
Student 1	36905626023
Student 2	36905626184
Student 3	36905625989
Student 4	36905626030
Student 5	36905626058

Workflow	Successful run
Integration	36905626020

6.6 Docker Compose deployment

The [3 October stack status](#) records 20 running containers: edge, F1's six, F2/F3/F5's three each, and F4's four. All 18 healthchecked containers are healthy; F1 runner and loader have no Docker healthcheck. AI-mode/MCP/RAG are separate running host processes. Shared home and all five feature pages return 200 in the operations record. Backends retain the generated host connection configuration; Compose defines no AI service.

Live service group	Running / healthy
Shared edge	1 / 1
Feature 1	6 / 4; runner/loader have no healthcheck
Features 2, 3 and 5	3 / 3 each
Feature 4	4 / 4
Host AI-mode, MCP, RAG	3 running; authenticated probe passes

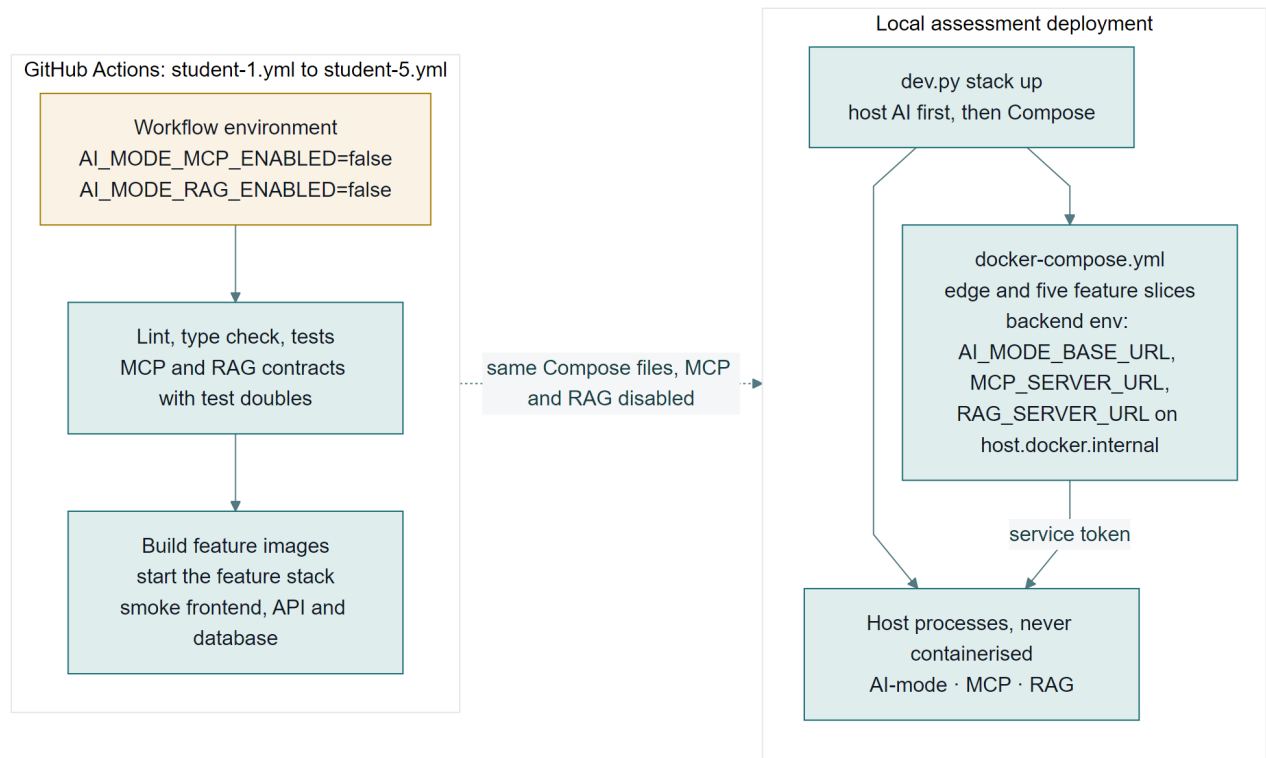


Figure 21 CI and local deployment

7 Integration summary

One Compose project and one host AI tier serve all five features. MCP reaches owned backend tools; RAG retrieves each registered corpus. The frontend scenarios cover structured current facts, cited guidance and insufficient context, while the operations record covers ordinary database-backed behaviour. These selected paths establish integration; the limitations below constrain broader claims. Activity history links runs to their feature and evidence.

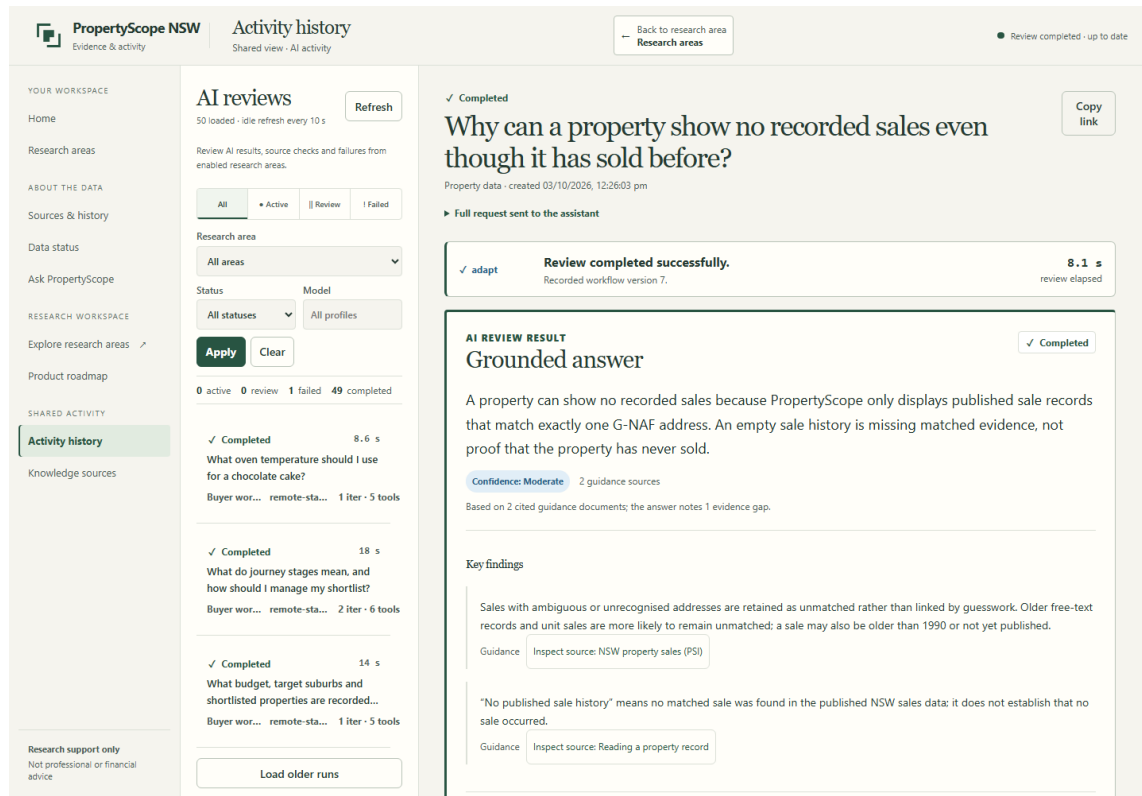


Figure 22 Cross-feature activity history

8 Known issues and limitations

- Corpora are project guidance, not official methodology; confidence does not prove entailment.
- Related but irrelevant passages can exceed the relevance floor. The F2 password example still refuses with zero citations; retrieval similarity alone does not establish a supported answer.
- Live operation requires a prepared host, embedding model and working provider access.
- The [accepted-data snapshot](#) records real G-NAF 5,190,134; PSI 7,402,643; BOCSAR 318,122; SEIFA 4,320. Schools (10) and fixture properties (3) remain seeded demonstrations. Unpublished candidates are not current data.
- F2 can retain old history when cases change, and old context during an active turn; F4's native generated-question renderer does not handle all grounded finding objects. These owner fixes are outside this Shared/F1 increment.
- F3 locality coverage is incomplete. F4's selected review uses source-attributed synthetic demonstration constraints/building records, not verified official coverage; missing evidence remains unknown.
- One exploratory F4 off-topic turn failed adaptation validation; the captured password question returned valid insufficient context. Selected successful scenarios do not establish all-question reliability.

- F5's offline AI-mode CI start needs owner review. The report does not certify every persistence table has ten rows or Q&A quality from repository checks. Video coverage still needs human review.

9 Contributions, repository and showcase

[Repository](#) and [showcase presentation](#) accompany this report. Matthew confirmed on 3 October that all five attended and participated in the 2 October showcase. Recorded participation: Matthew Shelton, Burhan Naeem, James Huang, Michael White and Derek Song. This confirms attendance; the marker assesses each student's demonstration and Q&A. The unauthenticated watch page loads; [player metadata](#) reports 575 seconds (9:35).

Student 1 Matthew Shelton

7 September: [1e109ac](#), shared MCP/RAG, host runtime, loop modes and F1 grounding. 19 September: [9bc4ee9](#), adoption kit. 26 September: [e7485af](#), retrieval/confidence evaluation. 3 October: [6aed375](#), verified MCP transport; this branch adds report guards, live capture and responsive Sources checks.

Student 2 Burhan Naeem

28 September: [a1b8577](#), F2 allowlist, corpus, capability tool and shared chat; [d18dfee](#), UI evidence/capture fixes. 1 October: [f019e41](#), assistant workspace and refreshed screenshots.

Student 3 James Huang

1 October: [da7556c2](#), map access to F1 data. 2 October: [b1181cd](#), tools, shared chat and corpus; [330e65f](#), guidance and published locality-context refinements.

Student 4 Michael White

1 October: [bbd112d](#), grounded allowlist/capability; [a6b7b7f](#), corpus; [e0720f4](#), shared assistant; [b4cf74a](#), CI tests; [7aa69f6](#), live evidence. [Owner log](#).

Student 5 Derek Song

1 October: [bb1e3b2](#), scoped assistant routes, corpus, cited UI and MCP/RAG evidence. [Owner log](#).

Contribution identities resolve in the selected baseline. Reproduce with `stack up, explicit corpus ingestion, ai probe, ai validate mcp, ai validate rag, capture_release1_screenshots.py, capture_release1_integration.py` and the full `check.py` gate. Build the single group-20.pdf with `build_release1_report.py --final --baseline <sha>`.